

DevOps Lab L3 Assignment Submission

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GitHub Repo: <https://github.com/Kranvagn4/hello-world-flask>

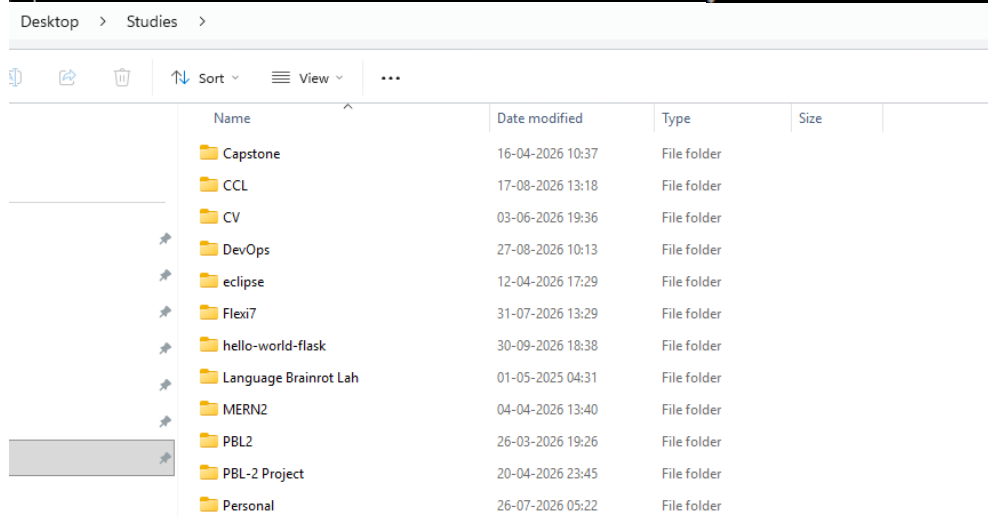
Part 1: Git Workflow and Collaboration

Task 1.1: Initialize the Git repository and push main:

I initialized a new Git repository for my “Hello World” Python Flask application, by adding the required Bash code to first create a directory, committed, and finally pushed it to the main branch.

Creation of the directory:

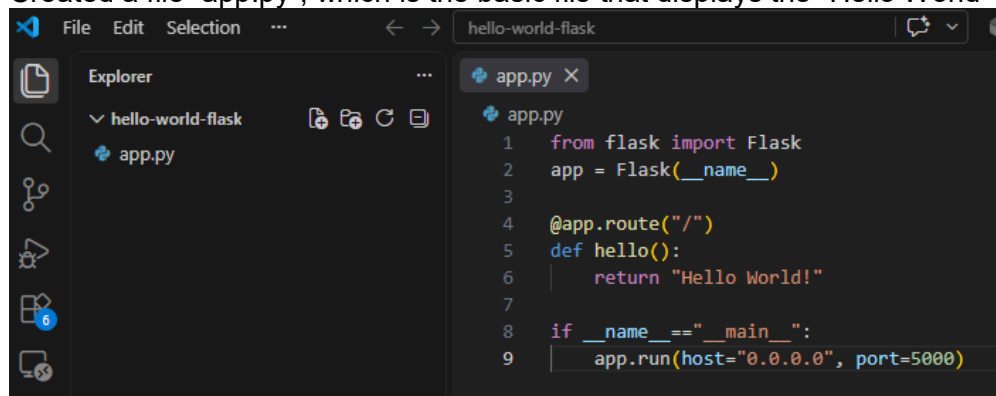
```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies $ mkdir hello-world-flask
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies $ cd hello-world-flask
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask $
```



Name	Date modified	Type	Size
Capstone	16-04-2026 10:37	File folder	
CCL	17-08-2026 13:18	File folder	
CV	03-06-2026 19:36	File folder	
DevOps	27-08-2026 10:13	File folder	
eclipse	12-04-2026 17:29	File folder	
Flexi7	31-07-2026 13:29	File folder	
hello-world-flask	30-09-2026 18:38	File folder	
Language Brainrot Lah	01-05-2025 04:31	File folder	
MERN2	04-04-2026 13:40	File folder	
PBL2	26-03-2026 19:26	File folder	
PBL-2 Project	20-04-2026 23:45	File folder	
Personal	26-07-2026 05:22	File folder	

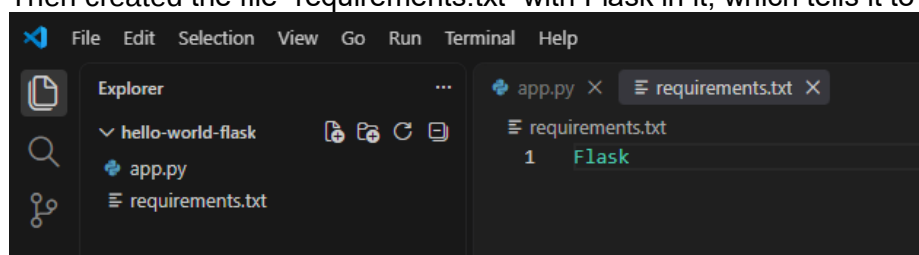
I then proceeded to create the Flask Application:

Created a file “app.py”, which is the basic file that displays the “Hello World” on port 5000.



```
1 from flask import Flask
2 app = Flask(__name__)
3
4 @app.route("/")
5 def hello():
6     return "Hello World!"
7
8 if __name__ == "__main__":
9     app.run(host="0.0.0.0", port=5000)
```

Then created the file “requirements.txt” with Flask in it, which tells it to download Flask.



```
1 Flask
```

Next, I initialized Git from our project folder, checked its status and added the files into the current directory.

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask
$ git init
Initialized empty Git repository in C:/Users/sanyu/OneDrive/Desktop/Studies/hello-world-flask/.git/

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.py
        requirements.txt

nothing added to commit but untracked files present (use "git add" to track)

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (master)
$ git add .

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (master)
$
```

Then, I checked the status once again.

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   app.py
        new file:   requirements.txt

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (master)
$
```

I then proceeded to create the initial commit by configuring my name and email, after setting the branch to main:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (master)
$ git config --global user.name "sanyu"

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (master)
$ git config --global user.email "sanyuktramolastrv@gmail.com"

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (master)
$ git commit -m "Initial Flask Hello World application"
[master (root-commit) 0bc13a1] Initial Flask Hello World application
 2 files changed, 10 insertions(+)
 create mode 100644 app.py
 create mode 100644 requirements.txt

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (master)
$ git branch -M main

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git branch
* main

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ |
```

Initial commit done successfully:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git status
On branch main
nothing to commit, working tree clean

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git log --oneline
0bc13a1 (HEAD -> main) Initial Flask Hello World application

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git branch
* main

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$
```

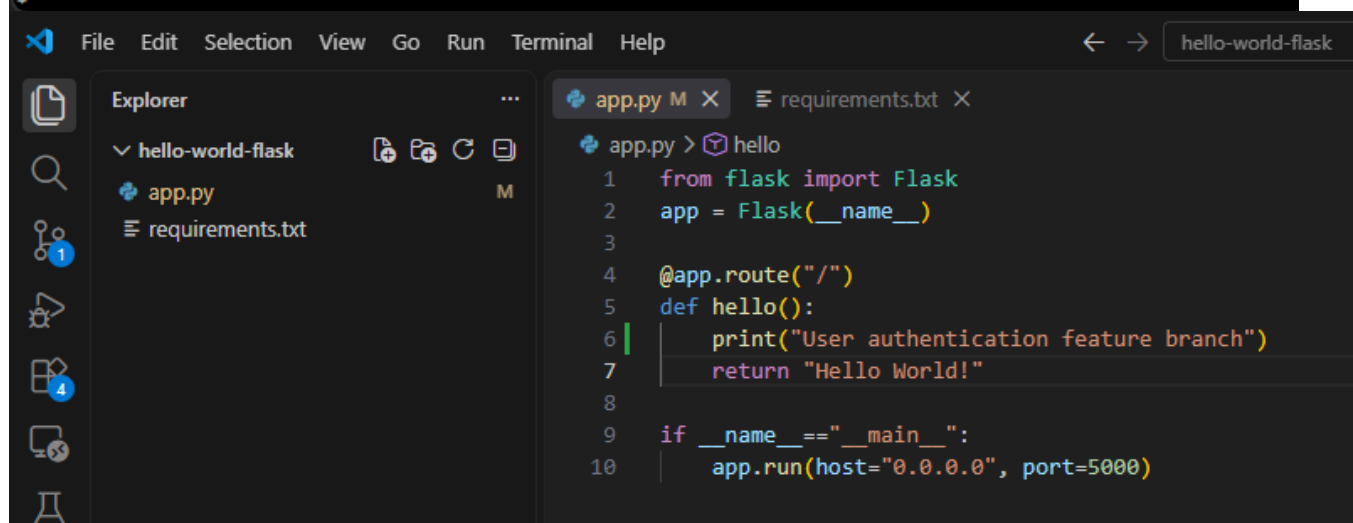
Task 1.2: Initialize the Git repository and push main:

I created a new branch in the repository called “feature/user-auth”, then proceeded to make changes into the flask application.

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git checkout -b feature/user-auth
Switched to a new branch 'feature/user-auth'

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git branch
* feature/user-auth
  main

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$
```



We can see the change by using the diff command, which tells us that we’ve added a print statement:

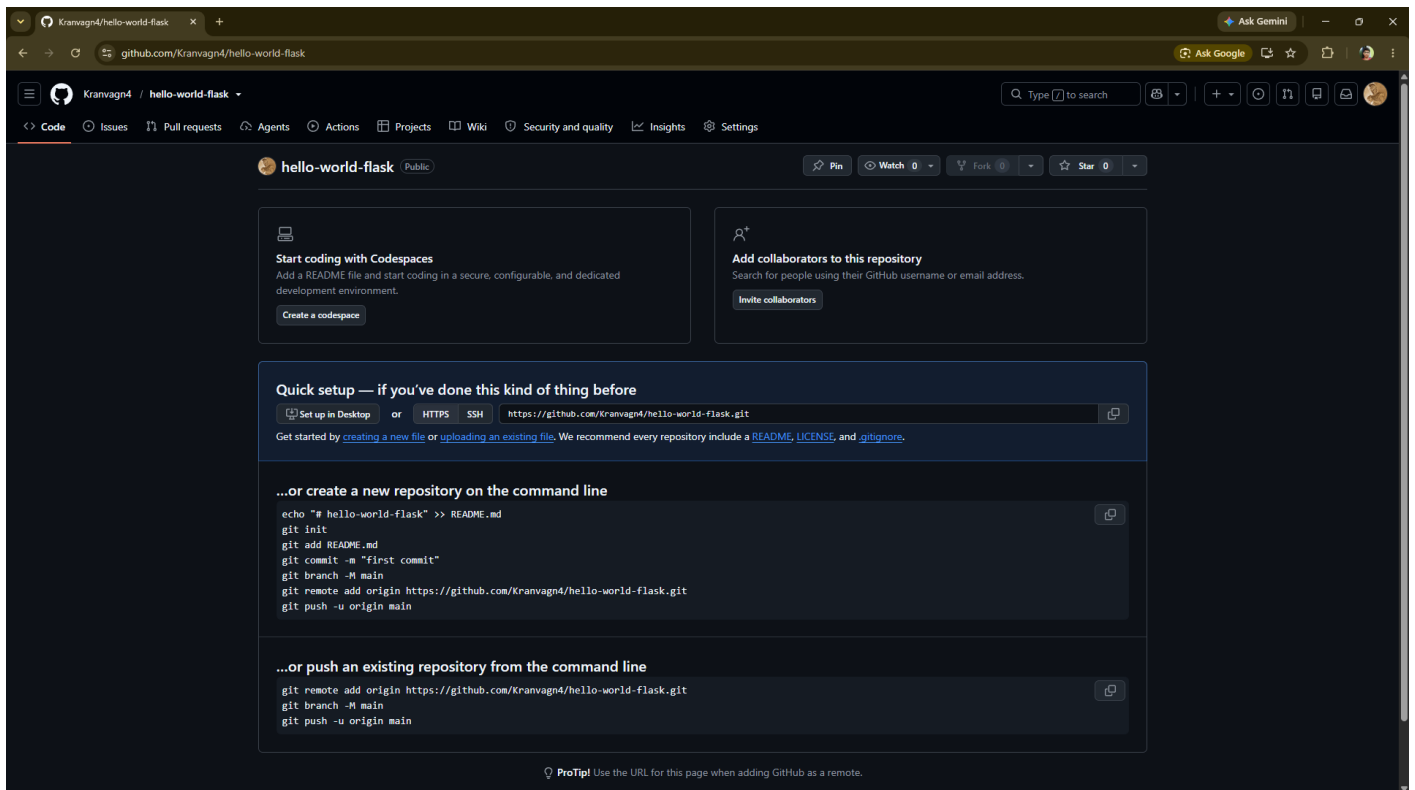
```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git diff
diff --git a/app.py b/app.py
index 825d97c..26ed33a 100644
--- a/app.py
+++ b/app.py
@@ -3,6 +3,7 @@ app = Flask(__name__)

 @app.route("/")
 def hello():
+    print("User authentication feature branch")
     return "Hello World!"

 if __name__ == "__main__":

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$
```

Now, we can go to Github and create our repository, and connect it to our local instance of the repo.



```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git remote add origin https://github.com/Kranvagn4/hello-world-flask

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ |

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git remote -v
origin https://github.com/Kranvagn4/hello-world-flask (fetch)
origin https://github.com/Kranvagn4/hello-world-flask (push)

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ |
```

Now, we push the local instance onto the Github repository:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 28 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 413 bytes | 413.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Kranvagn4/hello-world-flask
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$
```

Then, I did the same for the feature/user-auth branch:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git push -u origin feature/user-auth
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/user-auth' on GitHub by visiting:
remote:      https://github.com/Kranvagn4/hello-world-flask/pull/new/feature/user-auth
remote:
To https://github.com/Kranvagn4/hello-world-flask
 * [new branch]      feature/user-auth -> feature/user-auth
branch 'feature/user-auth' set up to track 'origin/feature/user-auth'.

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$
```

The repository has successfully been updated, containing both the branches:

The first screenshot shows the main view of the 'hello-world-flask' repository. It displays the 'main' branch with 2 branches and 0 tags. The file list includes 'app.py', 'requirements.txt', and a 'README' section with a prompt to 'Add a README'. The right sidebar shows repository statistics: 0 stars, 0 watching, 0 forks, and a language pie chart for Python at 100%.

The second screenshot shows the 'Branches' page for the same repository. It lists three branches: 'main' (updated 2 minutes ago, default branch), 'feature/user-auth' (updated 1 minute ago), and 'feature/user-auth' (active branch, updated 1 minute ago). Each branch entry includes a 'Check status' link and a 'Pull request' link.

Task 1.3: Simulating and Resolving a conflict:

For this task, we need to create an error which we will then resolve, which we will do by going to the main branch and creating a conflict between the main and the feature branch:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git checkout main
Switched to branch 'main'
M       app.py
Your branch is up to date with 'origin/main'.

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$
```

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git add app.py

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git commit -m "Modify hello message on main"
[main 5ab3547] Modify hello message on main
1 file changed, 1 insertion(+)

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 28 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 361 bytes | 361.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Kranvagn4/hello-world-flask
0bc13a1..5ab3547  main -> main

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$
```

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git checkout feature/user-auth
Switched to branch 'feature/user-auth'
Your branch is up to date with 'origin/feature/user-auth'.
```

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git status
On branch feature/user-auth
Your branch is up to date with 'origin/feature/user-auth'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ get diff
bash: get: command not found

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git diff
diff --git a/app.py b/app.py
index 825d97c..26ed33a 100644
--- a/app.py
+++ b/app.py
@@ -3,6 +3,7 @@ app = Flask(__name__)

@app.route("/")
def hello():
+   print("User authentication feature branch")
   return "Hello World!"

if __name__ == "__main__":

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git add app.py

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git commit -m "Add user authentication feature message"
[feature/user-auth 47bc3fe] Add user authentication feature message
1 file changed, 1 insertion(+)
```

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git push origin feature/user-auth
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 28 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 374 bytes | 374.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Kranvagn4/hello-world-flask
    Obc13a1..47bc3fe  feature/user-auth -> feature/user-auth

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ |
```

Now that we've successfully updated the code, we merge the branches:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (feature/user-auth)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git merge feature/user-auth
Auto-merging app.py
CONFLICT (content): Merge conflict in app.py
Automatic merge failed; fix conflicts and then commit the result.

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main|MERGING)
$
```

We can also see the conflict within app.py:

Then we resolve the conflict:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main|MERGING)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")
```

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main|MERGING)
$ git add app.py

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main|MERGING)
$ git commit -m "Resolve merge conflict between main and feature/user-auth"
[main 1ba7cff] Resolve merge conflict between main and feature/user-auth

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 28 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 414 bytes | 414.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Kranvagn4/hello-world-flask
 5ab3547..1ba7cff  main -> main

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$
```

Summary:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ git log --oneline --graph --all
* 1ba7cff (HEAD -> main, origin/main) Resolve merge conflict between main and feature/user-auth
| \
| * 47bc3fe (origin/feature/user-auth, feature/user-auth) Add user authentication feature message
* | 5ab3547 Modify hello message on main
| /
* 0bc13a1 Initial Flask Hello World application

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$
```

Final GitHub Repository:

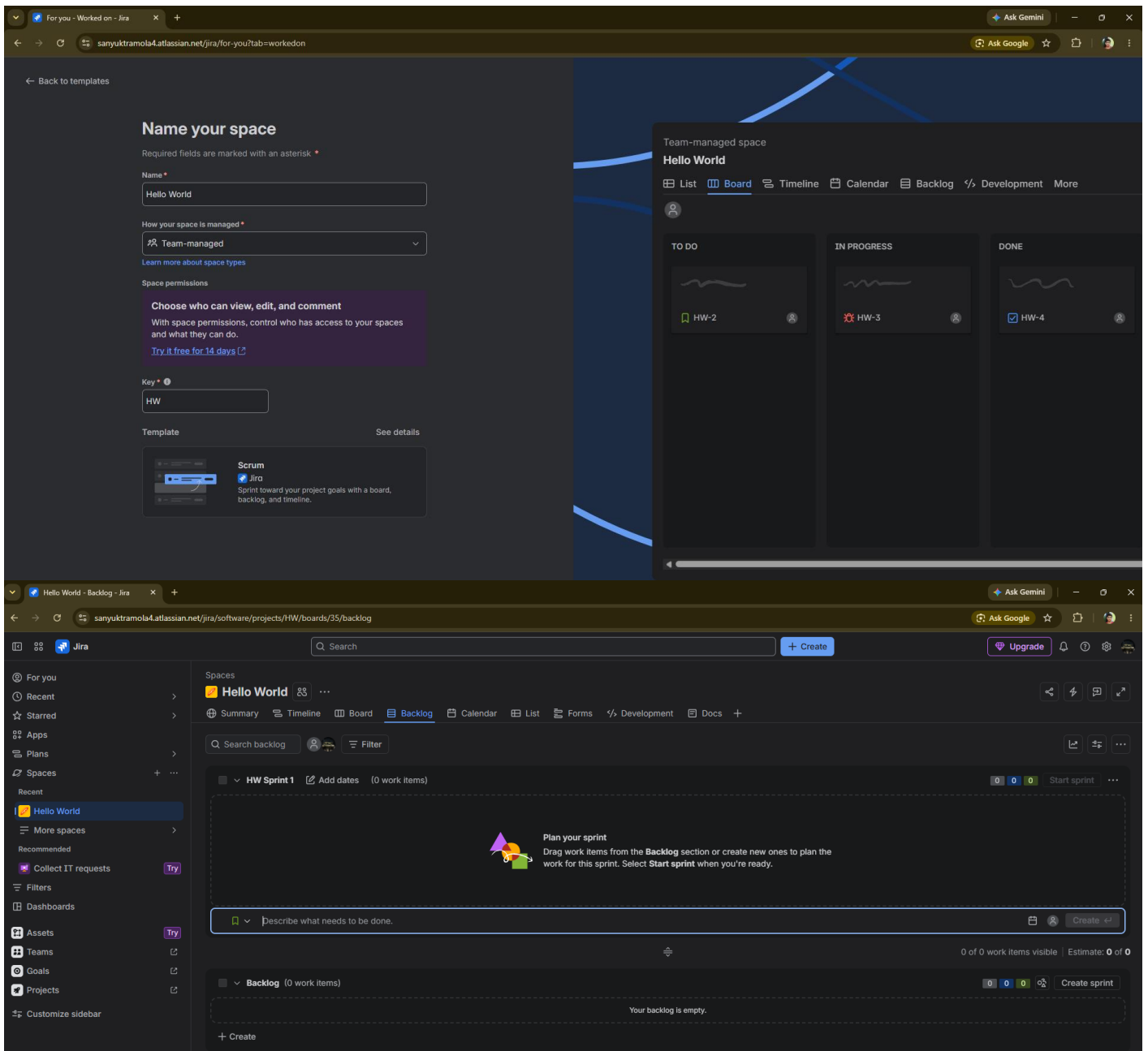
The first screenshot shows the main page of the GitHub repository 'hello-world-flask'. It displays the repository name, a search bar, and navigation tabs for Code, Issues, Pull requests, Agents, Actions, Projects, Wiki, Security and quality, Insights, and Settings. The repository has 2 branches and 0 tags. A list of commits is shown, including 'Resolve merge conflict between main and feature/user-auth' (1ba7cff), 'Add user authentication feature message' (47bc3fe), 'Modify hello message on main' (5ab3547), and 'Initial Flask Hello World application' (0bc13a1). The README section is visible, prompting the user to 'Add a README'. The right sidebar shows repository statistics: 0 stars, 0 watching, 0 forks, 0 releases, 0 packages, and 1 contributor (Kranvagn4). The language bar shows 100% Python.

The second screenshot shows the 'Commits' page for the 'main' branch. It displays a list of commits with their messages, commit hashes, and the user who made the commit. The commits are: 'Resolve merge conflict between main and feature/user-auth' (1ba7cff), 'Add user authentication feature message' (47bc3fe), 'Modify hello message on main' (5ab3547), and 'Initial Flask Hello World application' (0bc13a1).

Part 2: Jira Project and Issue tracking:

Task 2.1: Creating a Scrum project:

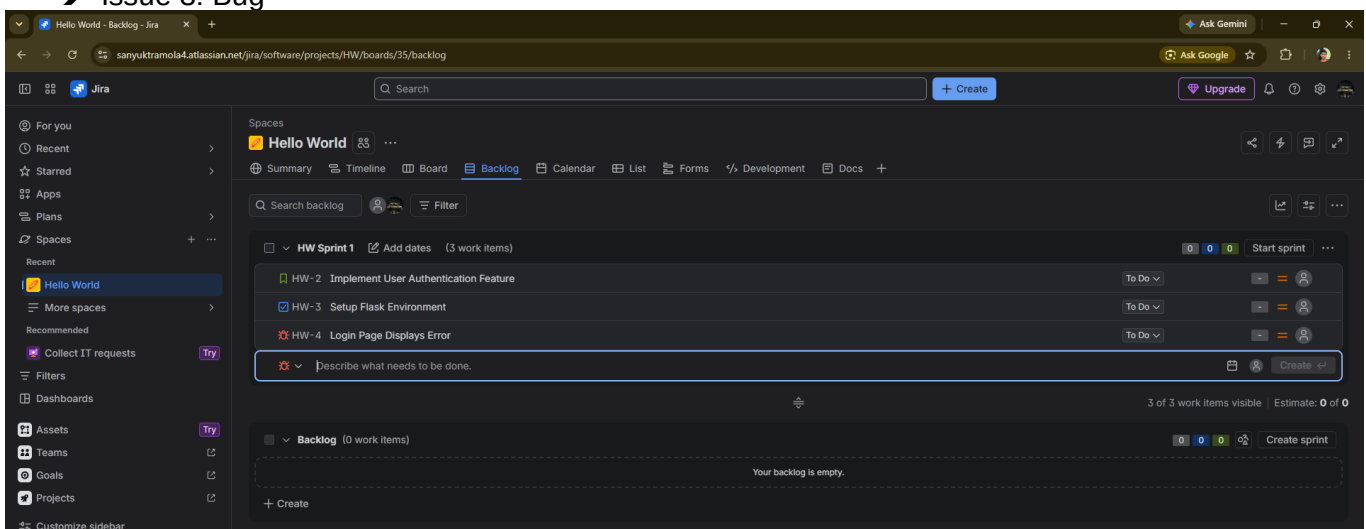
I logged onto my Atlassian account, then created a new space, scrum project named “Hello World”.



Task 2.2: Creating 3 issues onto the Scrum project:

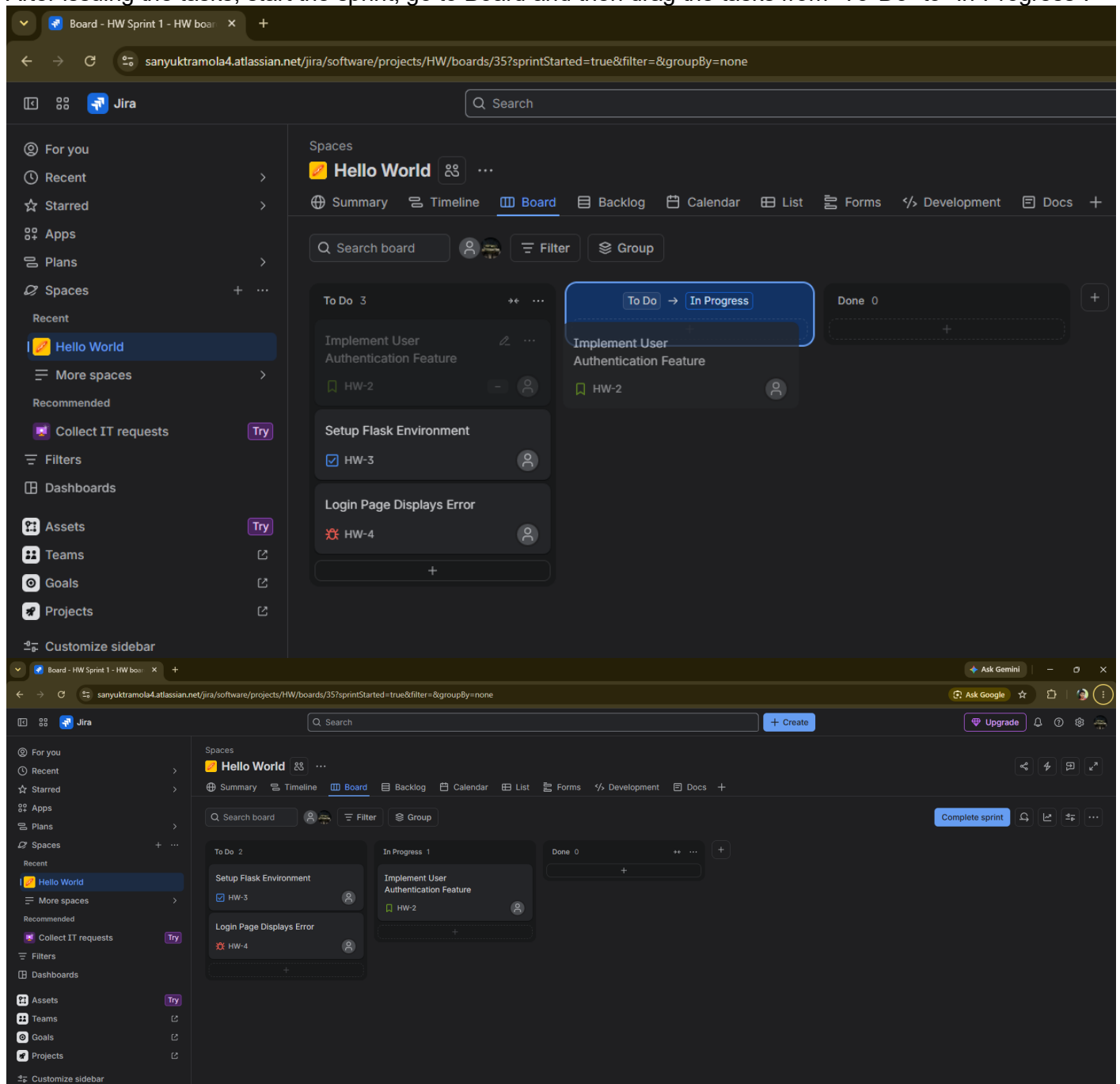
Now, we create 3 distinct issues:

- ➔ Issue 1: Story
- ➔ Issue 2: Task
- ➔ Issue 3: Bug



Task 2.3: Moving the tasks to “In-Progress”:

After issuing the tasks, start the sprint, go to Board and then drag the tasks from “To-Do” to “In-Progress”.



Part 3: Basic Containerization (Docker) and Jenkins:

Task 3.1: Creating the Dockerfile:

Inside our flask project, create a file called “Dockerfile” to run on Port 5000.

```
File Edit Selection View Go Run Terminal Help
hello-world-flask

Explorer
hello-world-flask
  app.py
  Dockerfile
  requirements.txt

Dockerfile U
Dockerfile > ...
1 FROM python:3.11-slim
2 WORKDIR /app
3 COPY requirements.txt .
4 RUN pip install --no-cache-dir -r requirements.txt
5 COPY app.py .
6 EXPOSE 5000
7 CMD ["python", "app.py"]

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ ls
Dockerfile app.py requirements.txt
```

Now, we build the docker image through Bash:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ docker build -t hello-world-flask .
#0 building with "desktop-linux" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 DONE 0.0s

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 205B done
#1 DONE 0.0s

#2 [internal] load metadata for docker.io/library/python:3.11-slim
#2 DONE 3.0s

#3 [internal] load .dockerignore
#3 transferring context: 2B done
#3 DONE 0.0s

#4 [internal] load build context
#4 transferring context: 347B done
#4 DONE 0.0s

#5 [1/5] FROM docker.io/library/python:3.11-slim@sha256:e41613d42d4891e4930f79523f93f81bbc7632584ec65e36ab055f41a800b41e
#5 resolve docker.io/library/python:3.11-slim@sha256:e41613d42d4891e4930f79523f93f81bbc7632584ec65e36ab055f41a800b41e 0.0s done
#5 sha256:0526d5e29bf341cb1941d46a6d4ab64fcde4fca3dc3adccfb4dbf5728b88b61e 0B / 249B 0.2s
#5 sha256:0526d5e29bf341cb1941d46a6d4ab64fcde4fca3dc3adccfb4dbf5728b88b61e 249B / 249B 0.3s done
#5 sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 0B / 14.45MB 0.2s
#5 sha256:41e7217c2e506d048f89ca0c5cfef848c05ef990900627c19f99938c4aae57f8 0B / 1.29MB 0.2s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 0B / 29.83MB 0.2s
#5 sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 1.05MB / 14.45MB 0.8s
#5 sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 4.19MB / 14.45MB 0.9s
#5 sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 6.29MB / 14.45MB 1.1s
#5 sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 8.39MB / 14.45MB 1.2s
#5 sha256:41e7217c2e506d048f89ca0c5cfef848c05ef990900627c19f99938c4aae57f8 1.29MB / 1.29MB 1.1s done
#5 sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 11.53MB / 14.45MB 1.4s
#5 sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 12.58MB / 14.45MB 1.5s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 2.10MB / 29.83MB 1.4s
#5 sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 13.63MB / 14.45MB 1.7s
#5 sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 14.45MB / 14.45MB 1.7s done
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 5.24MB / 29.83MB 1.8s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 7.34MB / 29.83MB 2.1s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 9.44MB / 29.83MB 2.4s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 11.53MB / 29.83MB 2.6s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 14.68MB / 29.83MB 2.9s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 16.78MB / 29.83MB 3.0s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 18.87MB / 29.83MB 3.2s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 20.97MB / 29.83MB 3.3s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 25.17MB / 29.83MB 3.6s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 28.31MB / 29.83MB 3.8s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 29.83MB / 29.83MB 3.9s
#5 sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 29.83MB / 29.83MB 3.9s done
#5 extracting sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8
#5 extracting sha256:6b37362b3da78869050b894b799ad4df04f1f3b52774087db0d81151570244c8 0.5s done
#5 DONE 4.5s
```

```

#5 [1/5] FROM docker.io/library/python:3.11-slim@sha256:e41613d42d4891e4930f79523f93f81bbc7632584ec65e36ab055f41a800b41e
#5 extracting sha256:41e7217c2e506d048f89ca0c5cfef848c05ef990900627c19f99938c4aae57f8 0.1s done
#5 extracting sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138
#5 extracting sha256:58abdd9670ca8c69d03432ce188995dee67f0979357caf83866fe1ef2ec37138 0.3s done
#5 DONE 4.9s

#5 [1/5] FROM docker.io/library/python:3.11-slim@sha256:e41613d42d4891e4930f79523f93f81bbc7632584ec65e36ab055f41a800b41e
#5 extracting sha256:0526d5e29b7341cb1941d46a6d4ab64fcde4fca3dc3adccfb4dbf5728b88b61e done
#5 DONE 4.9s

#6 [2/5] WORKDIR /app
#6 DONE 0.2s

#7 [3/5] COPY requirements.txt .
#7 DONE 0.0s

#8 [4/5] RUN pip install --no-cache-dir -r requirements.txt
#8 1.137 Collecting Flask (from -r requirements.txt (line 1))
#8 1.292   Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
#8 1.325 Collecting blinker>=1.9.0 (from Flask->-r requirements.txt (line 1))
#8 1.335   Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
#8 1.369 Collecting click>=8.1.3 (from Flask->-r requirements.txt (line 1))
#8 1.393   Downloading click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
#8 1.452 Collecting itsdangerous>=2.2.0 (from Flask->-r requirements.txt (line 1))
#8 1.468   Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
#8 1.513 Collecting jinja2>=3.1.2 (from Flask->-r requirements.txt (line 1))
#8 1.528   Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
#8 1.609 Collecting markupsafe>=2.1.1 (from Flask->-r requirements.txt (line 1))
#8 1.626   Downloading markupsafe-3.0.3-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metad
ata (2.7 kB)
#8 1.664 Collecting werkzeug>=3.1.0 (from Flask->-r requirements.txt (line 1))
#8 1.678   Downloading werkzeug-3.1.9-py3-none-any.whl.metadata (4.1 kB)
#8 1.696   Downloading flask-3.1.3-py3-none-any.whl (103 kB)
#8 1.740   ----- 103.4/103.4 kB 2.6 MB/s eta 0:00:00
#8 1.765   Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
#8 1.778   Downloading click-8.5.0-py3-none-any.whl (125 kB)
#8 1.787   ----- 125.3/125.3 kB 20.4 MB/s eta 0:00:00
#8 1.803   Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
#8 1.817   Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
#8 1.828   ----- 134.9/134.9 kB 14.3 MB/s eta 0:00:00
#8 1.842   Downloading markupsafe-3.0.3-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
#8 1.856   Downloading werkzeug-3.1.9-py3-none-any.whl (228 kB)
#8 1.874   ----- 228.7/228.7 kB 15.1 MB/s eta 0:00:00
#8 1.906 Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeug, jinja2, Flask
#8 2.139 Successfully installed Flask-3.1.3 blinker-1.9.0 click-8.5.0 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeu
g-3.1.9
#8 2.140 WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system pa
ckage manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
#8 2.222
#8 2.222 [notice] A new release of pip is available: 24.0 -> 26.2.1
#8 2.222 [notice] To update, run: pip install --upgrade pip
#8 DONE 2.3s

#9 [5/5] COPY app.py .
#9 DONE 0.0s

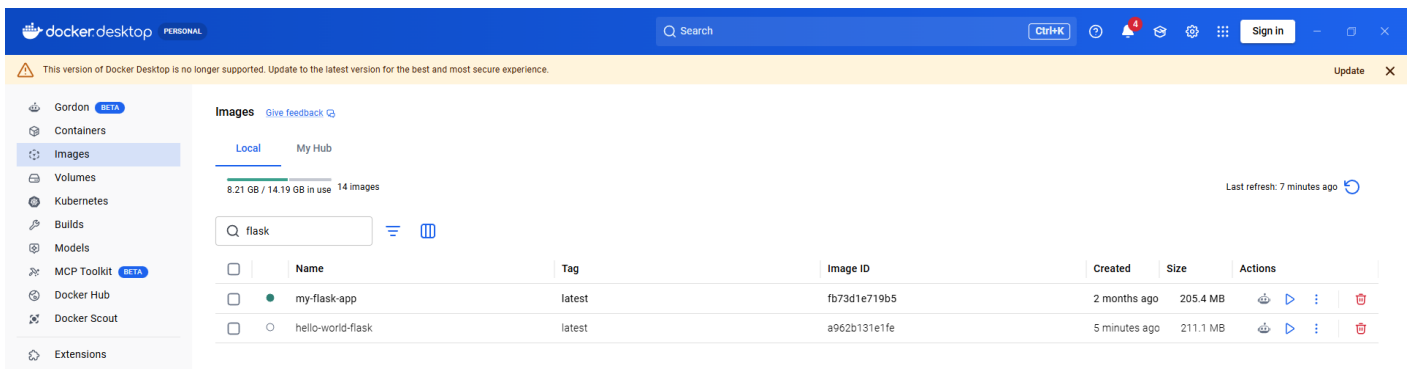
#10 exporting to image
#10 exporting layers
#10 exporting layers 0.5s done
#10 exporting manifest sha256:d58cc70895d6bc2e2690b7b315875e10b15185fdc0597db36f359fb744dcc0b1 0.0s done
#10 exporting config sha256:329b155cf9eadbf74b88d3d4f7bef5132aa33d76098f4ff4916b4680b68a134a 0.0s done
#10 exporting attestation manifest sha256:85a0d716733e0e61f2e35a0ac6c78ab8465c839845741152023adb849c1217bf 0.0s done
#10 exporting manifest list sha256:a962b131e1feece41ac5e05cd38846a44d4ca9563a051cea849c0e6d8feff94 done
#10 naming to docker.io/library/hello-world-flask:latest done
#10 unpacking to docker.io/library/hello-world-flask:latest
#10 unpacking to docker.io/library/hello-world-flask:latest 0.2s done
#10 DONE 0.7s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/woi4afu7amcrjaw9r2y7gcmh0

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$

```

We can see the image built on Docker:



```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ docker images
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
IMAGE                                ID                                DISK USAGE    CONTENT SIZE  EXTRA
alpine/openssl:latest                9eed1eb6f96f                    14.2MB        4.22MB
chromadb/chroma:latest               1e0b73a187a2                    826MB        176MB        U
docker.n8n.io/n8nio/n8n:latest       882b126a8ddd                    2.48GB        387MB        U
docker/desktop-cloud-provider-kind:v0.5.0 4ad59ce20658                    595MB        162MB        U
docker/desktop-containerd-registry-mirror:v0.0.3 179a9ea3ae47                    50.3MB        16.1MB        U
docker/welcome-to-docker:latest      c4d56c24da4f                    22.2MB        6.03MB
envoyproxy/envoy:v1.36.4            08acc53db6f4                    248MB        62.1MB
ghcr.io/open-webui/open-webui:main   6a773e5c3a24                    7.16GB        1.84GB        U
hello-world-flask:latest             a962b131e1fe                    211MB        51.3MB
jenkins/jenkins:lts-jdk17           cce1eeb79902                    785MB        280MB
kindest/node:v1.34.3                08497ee19eac                    1.35GB        395MB        U
my-flask-app:latest                  fb73d1e719b5                    205MB        49.5MB        U
my-node-app:latest                   6129fa01796d                    194MB        47MB
n8nio/n8n:latest                     882b126a8ddd                    2.48GB        387MB        U
openquantumsafe/openssl:latest       543fb00ce31d                    79.3MB        17.8MB
```

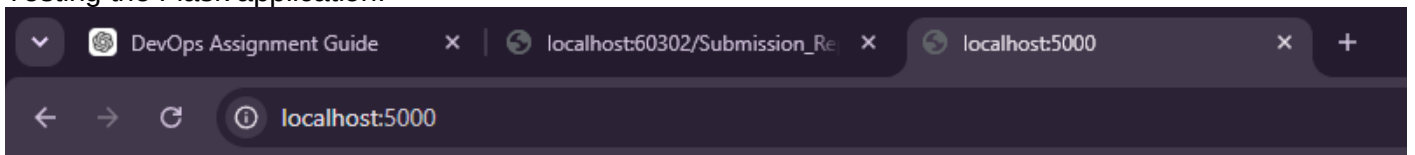
Now, we run the container, and check the run:

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ docker run -d -p 5000:5000 --name hello-flask hello-world-flask
21c17a84c09c5007f1b156de2c667095dee789feb80c9752200b934f26dcaa89

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$ docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS
21c17a84c09c   hello-world-flask "python app.py"         28 seconds ago Up 27 seconds 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
hello-flask

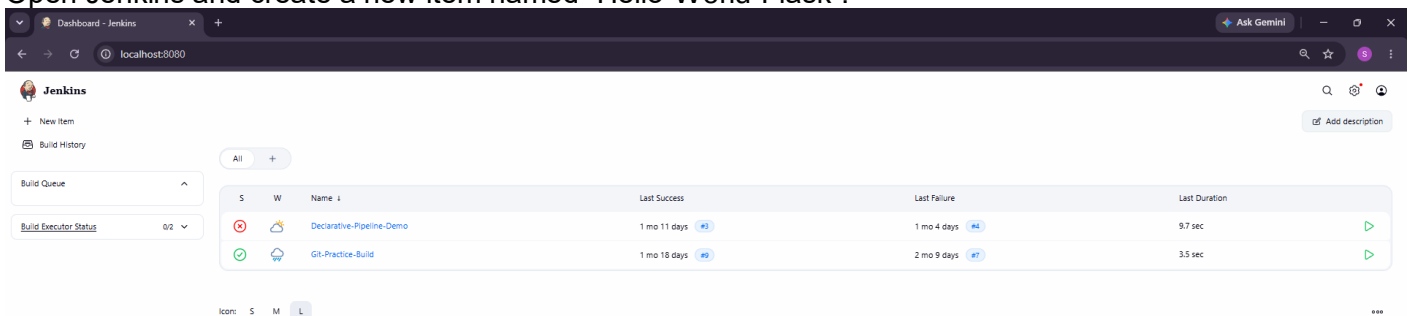
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/hello-world-flask (main)
$
```

Testing the Flask application:



Task 3.2: Jenkins Freestyle Project:

Open Jenkins and create a new item named "Hello-World-Flask".



New Item

Enter an item name

Hello-World-Flask

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Configure the Source Code Management as Git, and enter the GitHub repo link:

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☒ None

☐ Git ?

Then add the branch to build as main:

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/Kranvagn4/hello-world-flask

Credentials ?

- none -

+ Add

Advanced ▾

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Now, add the build steps:

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

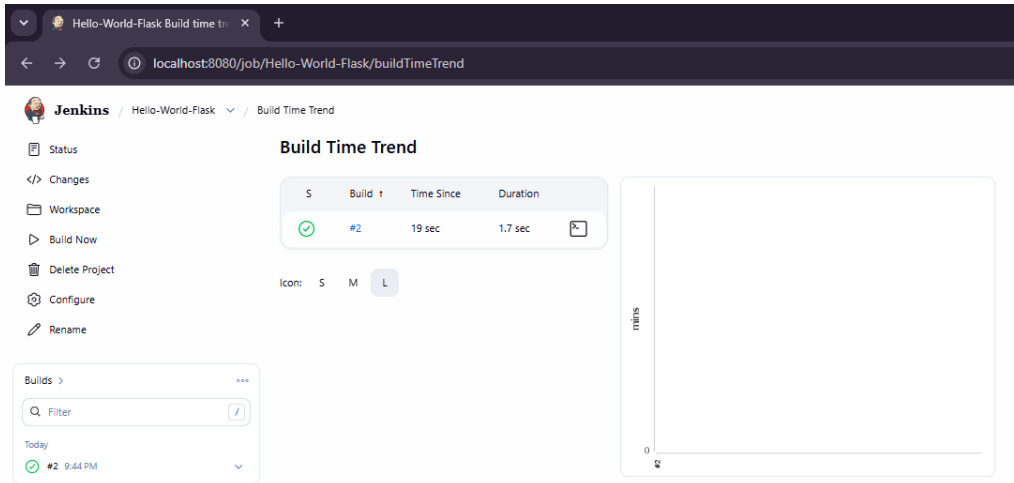
Command

See the list of available environment variables

```
echo.
echo Files in Workspace:
dir
echo.
echo ===== Application Files =====
dir /s /b
```

Advanced ▾

Build the project:



Build #2 (Sep 30, 2026, 9:44:37 PM)

Started by user Sanyukt Ramola

Revision: 1ba7cff4ba369856182f7f14d263beab33c30d80
Repository: <https://github.com/Kranvagn4/hello-world-flask>

- refs/remotes/origin/main

No changes.

Console Output:

Started by user Sanyukt Ramola

Running as SYSTEM

Building in workspace C:\ProgramData\Jenkins\workspace\Hello-World-Flask

The recommended git tool is: NONE

No credentials specified

```
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\workspace\Hello-World-Flask\.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/Kranvagn4/hello-world-flask # timeout=10
Fetching upstream changes from https://github.com/Kranvagn4/hello-world-flask
> git.exe --version # timeout=10
> git --version # 'git version 2.46.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/Kranvagn4/hello-world-flask +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 1ba7cff4ba369856182f7f14d263beab33c30d80 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 1ba7cff4ba369856182f7f14d263beab33c30d80 # timeout=10
Commit message: "Resolve merge conflict between main and feature/user-auth"
First time build. Skipping changelog.
[Hello-World-Flask] $ cmd /c call C:\WINDOWS\TEMP\jenkins17067701165544675577.bat

C:\ProgramData\Jenkins\workspace\Hello-World-Flask>echo ===== Jenkins Workspace Contents =====
===== Jenkins Workspace Contents =====

C:\ProgramData\Jenkins\workspace\Hello-World-Flask>echo.

C:\ProgramData\Jenkins\workspace\Hello-World-Flask>echo Current Directory:
Current Directory:

C:\ProgramData\Jenkins\workspace\Hello-World-Flask>cd
C:\ProgramData\Jenkins\workspace\Hello-World-Flask
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask>echo.

C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask>echo Files in Workspace:
Files in Workspace:

C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask>dir
Volume in drive C is Windows-SSD
Volume Serial Number is AA46-898A

Directory of C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask

30-09-2026  21:42    <DIR>        .
30-09-2026  21:42    <DIR>        ..
30-09-2026  21:42                262 app.py
30-09-2026  21:42                5 requirements.txt
                2 File(s)                267 bytes
                2 Dir(s)  55,308,066,816 bytes free

C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask>echo.

C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask>echo ==== Application Files ====
==== Application Files ====

C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask>dir /s /b
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\app.py
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\requirements.txt
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\config
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\description
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\FETCH_HEAD
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\HEAD
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\index
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\info
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\logs
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\refs
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\applypatch-msg.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\commit-msg.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\fsmonitor-watchman.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\post-update.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\pre-applypatch.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\pre-commit.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\pre-merge-commit.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\pre-push.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\pre-rebase.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\pre-receive.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\prepare-commit-msg.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\push-to-checkout.sample
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\sendemail-validate.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\hooks\update.sample
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\info\exclude
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\logs\HEAD
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\logs\refs
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\logs\refs\remotes
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\logs\refs\remotes\origin
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\logs\refs\remotes\origin\feature
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\logs\refs\remotes\origin\main
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\logs\refs\remotes\origin\feature\user-auth
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\0b
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\1b
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\1c
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\20
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\26
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\47
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\57
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\5a
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\77
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\82
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\98
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\9a
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\de
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\info
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\pack
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\0b\c13a19dd5908c720a7566be60a7a1721681197
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\1b\7c7ff4ba369856182f7f14d263beab33c30d00
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\1c\1fa93bd08c74d0453a1f2a810aa4a8cfed26a
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\20\77213c37cc027f3b2ee884c940c2f41d7adea7
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\26\ed33a21f47ec2547a541130be77dbefdcbe5a4
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\47\bc3fe51ed1de5c34f2d8b7d3de244b3298e6d2
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\57\9161e847f561701dd072bba21e70fc31d280ab
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\5a\b3547c4823ff2e961a7d26f2fef199abf026c8
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\77\9c859bda3d6791fdb3e33d99b0bf816dd56f8e
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\82\5d97cb79a65a871ceb364e74491c0dc62334b
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\98\5dded299919038469d2a94f2ae0419f23fa704
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\9a\2f6e314d3c90dd118ceb5a41a46d1a9df83262
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\objects\de\ed59dd47d7de892ef2606915026c21617ecb2
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\refs\heads
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\refs\remotes
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\refs\tags
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\refs\remotes\origin
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\refs\remotes\origin\feature
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\refs\remotes\origin\main
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask\.git\refs\remotes\origin\feature\user-auth

C:\ProgramData\Jenkins\.jenkins\workspace\Hello-world-Flask>exit 0
Finished: SUCCESS
```